Tutorial 3

Systems of linear equations, logic and elementary matrices

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1. For each of the following statements say whether it is true or false, and give a reason for your answer.

   (a) If $m = n$, then for every $A \in M_{m\times n}(\mathbb{R})$ and every m-vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.

   Solution: Note that "a system has a unique solution" means that the system has exactly one solution, i.e., the system has a solution, but not more than one solution.
   The statement is false.
   Take for instance A $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Then $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions (hence not a unique solution).

   Or take A $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$. Then $A\mathbf{x} = \mathbf{b}$ is inconsistent, i.e., has no solution (and hence not a unique solution).
   There are many other counterexamples.

   (b) If $m > n$, then for every $A \in M_{m\times n}(\mathbb{R})$ and every m-vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has at least one solution.

   Solution: False. Take for instance A $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.
   There are many other counterexamples.

   (c) If $m > n$, then for every $A \in M_{m\times n}(\mathbb{R})$ and every m-vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has no solution.

   Solution: False. Take for instance A $= \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.
   There are many other counterexamples.

   (d) If $m < n$, then for every $A \in M_{m\times n}(\mathbb{R})$ and every m-vector $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.

Solution: False. Take for instance A $= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

There are many other counterexamples.

2. Let U be a row echelon form of the augmented matrix of the system $A\mathbf{x} = \mathbf{b}$, where $A \in M_{m\times n}(\mathbb{R})$. Determine whether the following statements are true or false, and provide a reason for your answer.

   (a) If U contains a row of 0's, the system $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions.

   Solution: False. Take for instance U $= \left(\begin{array}{cc|c} 0 & 0 & 1 \\ 0 & 0 & 0 \end{array}\right)$. The corresponding system $A\mathbf{x} = \mathbf{b}$ is inconsistent.

   (b) If there are $n+1$ pivots in U, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.

   Solution: False. Take for instance U $= \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)$. The corresponding system $A\mathbf{x} = \mathbf{b}$ is inconsistent.

   (c) If U does not contain a row of 0's, it is not in row echelon form.

   Solution: False. Take for instance U $= \left(\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & 7 \end{array}\right)$. This is a matrix in row echelon form but does not contain a row of 0's.

3. Consider the following system of linear equations:

$$\begin{aligned} 2x_2 + 4x_3 &= 18 \\ -x_1 - x_2 + 2x_3 &= -4 \\ 2x_1 + 18x_3 &= 16 \\ 2x_1 - 4x_2 + 10x_3 &= -20 \end{aligned}$$

   (a) Write down the augmented matrix of this system. Is this matrix in row echelon form? Give a reason for your answer.

   Solution:

$$\left(\begin{array}{ccc|c} 0 & 2 & 4 & 18 \\ -1 & -1 & 2 & -4 \\ 2 & 0 & 18 & 16 \\ 2 & -4 & 10 & -20 \end{array}\right)$$

   Not in row echelon form; there are many reasons for this. Find at least two!

(b) Use **the algorithm for Gaussian elimination that we've given you in the notes** to reduce the augmented matrix to row echelon form. Follow the steps in the algorithm **as given in the notes carefully, without taking short-cuts**. Stop when the algorithm tells you to do so. This should go *quickly*! (As you can see, it is important to consult your notes when doing the tutorial...)

Solution:

$$\left(\begin{array}{ccc|c} 0 & 2 & 4 & 18 \\ -1 & -1 & 2 & -4 \\ 2 & 0 & 18 & 16 \\ 2 & -4 & 10 & -20 \end{array}\right) \xrightarrow{R_2 \longleftrightarrow R_1} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 2 & 0 & 18 & 16 \\ 2 & -4 & 10 & -20 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_1} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 0 & -2 & 22 & 8 \\ 2 & -4 & 10 & -20 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 + 2R_1} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 0 & -2 & 22 & 8 \\ 0 & -6 & 14 & -28 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_2} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 0 & 0 & 26 & 26 \\ 0 & -6 & 14 & -28 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 + 3R_2} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 0 & 0 & 26 & 26 \\ 0 & 0 & 26 & 26 \end{array}\right)$$

$$\xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} -1 & -1 & 2 & -4 \\ 0 & 2 & 4 & 18 \\ 0 & 0 & 26 & 26 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(c) Does this system have a solution? If so, write down the solution set of the system.

Solution: Read off the solutions: $26x_3 = 26$, so $x_3 = 1$. Then $2x_2 = -4x_3 + 18 = -4(1) + 18 = 14$, so $x_2 = 7$. Then $-x_1 = x_2 - 2x_2 - 4 = 1$, so $x_1 = -1$. The solution set is $\{(-1, 7, 1)\}$.

(d) How would you now check your solution is correct? Do it!

Solution: Substitute them back into the original system of equations to see if all the equations hold.

In the next two questions you have to use some logic. Remember that if you believe a statement is *true*, you have to give a *proof* to show it is true; if you believe it is *false*, a *counter-example* is all that is needed to *disprove* it.

4. The statement

   For every x, $P(x) \Rightarrow Q(x)$

   has as its **converse** the statement

   For every x, $Q(x) \Rightarrow P(x)$.

   A statement and its converse are **NOT** logically equivalent. They mean quite different things.

   Now consider the statement:

   Every upper triangular matrix is a square matrix in row echelon form.

   (a) Is this statement true or false? Prove it or disprove it.

   Solution: False. A possible counter-example is the matrix $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$; it is upper triangular but not in row echelon form.

   (b) Write down the *converse* of this statement. Is the converse true or false? Prove it or disprove it.

   Solution: The converse is

   Every square matrix in row echelon form is upper triangular.

   The statement is true. Write out a complete argument yourself. Keep in mind that a column may or may not contain a pivot; if there is a pivot, it will be on or above the diagonal, and all entries in the column below the pivot must be 0. Make sure that you know why all of this is the case.

5. The statement

   For every x, $P(x) \Rightarrow Q(x)$

   is logically equivalent to the statement

For every x, $\neg Q(x) \Rightarrow \neg P(x)$. (The symbol $\neg P$ means "not P" or "it is not true that P".)

The second statement is called the **contrapositive** of the first. A statement and its contrapositive **are** logically equivalent. They have exactly the same meaning.

Consider the statement:

For all real numbers k and all matrices A, if $kA = \mathbf{0}$ then $k = 0$ or $A = \mathbf{0}$.

Write down and then prove its contrapositive.

> **Solution:** The contrapositive is
>
> For all real numbers k and all matrices A, if $k \neq 0$ and $A \neq \mathbf{0}$ then $kA \neq \mathbf{0}$.
>
> *Proof:* If $A \neq \mathbf{0}$, then it has at least one non-zero entry, say a_{ij}. Then kA has entry ka_{ij}, which must be non-zero, because $k \neq 0$ and $a_{ij} \neq 0$. So $kA \neq \mathbf{0}$.

6. Which of the following matrices are elementary? Give reasons for your answers.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & -7 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

> **Solution:** A is elementary (it is obtained from I_3 by subtracting 4 times row 1 from row 3).
> B is not elementary (it takes **two** elementary row operations to change I_3 into A).
> C is elementary (it is obtained from I_3 by interchanging row 1 and 2).
> D is not elementary (No elementary row operation can change I_3 into D — multiplying a row by zero is not an elementary row operation).

7. For each of the following two matrices, use **the definitions** to decide whether it is singular or non-singular:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Solution: Please don't use determinants to solve this question! Solve this using only the definition of singularity.

A is non-singular because the equation $A\mathbf{x} = \mathbf{0}$ has $\mathbf{x} = \mathbf{0}$ as its only solution. Try to see this without much calculation: First, the last row of A tells us that $x_3 = 0$. Then row 2 implies that $x_1 = 0$ and finally, row 1 of A implies that $x_2 = 0$.

B is singular because the equation $B\mathbf{x} = \mathbf{0}$ has infinitely many solutions. You can see this because all entries in column 2 are zero (implying that x_2 can take any value (and $x_1 = x_3 = 0$)).

8. Consider the statement:
For every $n \times n$ matrix A, if $A^2 = \mathbf{0}$, then $A + I_n$ is invertible. $(*)$
   (a) Write down the converse of the statement $(*)$.
   (b) Write down the contrapositive of the statement $(*)$.
   (c) Prove or disprove the statement $(*)$.
   (d) Prove or disprove the converse of the statement $(*)$.

Solution:

(a) For every $n \times n$ matrix A, if $A + I_n$ is invertible, then $A^2 = \mathbf{0}$.

(b) For every $n \times n$ matrix A, if $A + I_n$ is not invertible, then $A^2 \neq \mathbf{0}$.

(c) The statement is true.
If $A^2 = \mathbf{0}$, then

$$(A + I_n)(I_n - A) = A - A^2 + I_n - A = I_n$$

and

$$(I_n - A)(A + I_n) = A + I_n - A^2 - A = I_n.$$

Hence by definition, $A + I_n$ is invertible and $(A + I_n)^{-1} = I_n - A$.

(d) The statement is false. As a counterexample, take $A = I_n$. Then $A + I_n = 2I_n$, which is clearly invertible, but $A^2 = I_n \neq \mathbf{0}$.